

Midterm

Phil 305 — Propositional and Modal Logic

Part 1: S5

Use the simplified rules for **S5** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **S5**.

1. $\Box(\Box p \rightarrow q) \vee \Box(\Box q \rightarrow p)$
2. $\Diamond\Box p \rightarrow \Box\Diamond p$
3. $(\neg p \vee \neg\Box\Box p) \rightarrow \neg\Box p$

Part 2: K

Use the rules for **K** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **K**.

4. $(\Diamond p \wedge \Diamond q) \rightarrow \Box(p \vee q)$
5. $\Diamond(p \rightarrow p) \vee \Diamond\neg(q \rightarrow q)$
6. $(p \wedge \neg\neg\Box q) \rightarrow (\neg\neg p \wedge \Box q)$

Part 3: S4

Use the rules for **S4** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **S4**. Remember **S4** is the logic **KT4**, so it has the rules for **T** and **4** as well as **K**.

7. $\Box(p \rightarrow q) \rightarrow (\Diamond\Diamond p \rightarrow \Diamond q)$
8. $(p \wedge \Diamond\Box p) \rightarrow \Box p$

1. $\Box(\Box p \rightarrow q) \vee \Box(\Box q \rightarrow p)$ in **S5**

2. $\Diamond \Box p \rightarrow \Box \Diamond p$ in S5

3. $(\neg p \vee \neg \Box \Box p) \rightarrow \neg \Box p$ in **S5**

4. $(\Diamond p \wedge \Diamond q) \rightarrow \Box(p \vee q)$ **in K**

5. $\Diamond(p \rightarrow p) \vee \Diamond\neg(q \rightarrow q)$ in **K**

6. $(p \wedge \neg\neg\Box q) \rightarrow (\neg\neg p \wedge \Box q)$ **in K**

7. $\Box(p \rightarrow q) \rightarrow (\Diamond\Diamond p \rightarrow \Diamond q)$ in **S4**

8. $(p \wedge \Diamond \Box p) \rightarrow \Box p$ **in S4**